Job Market Analysis and Recommendation System

Project report based on the supplied Upwork job-posting snapshot

Dataset size: 244,828 rows × 8 original columns
Observed dates: 2023-11-02 09:22:02 to 2024-03-24 14:16:47
Countries: 213 distinct country labels including Unknown

1. Executive Summary

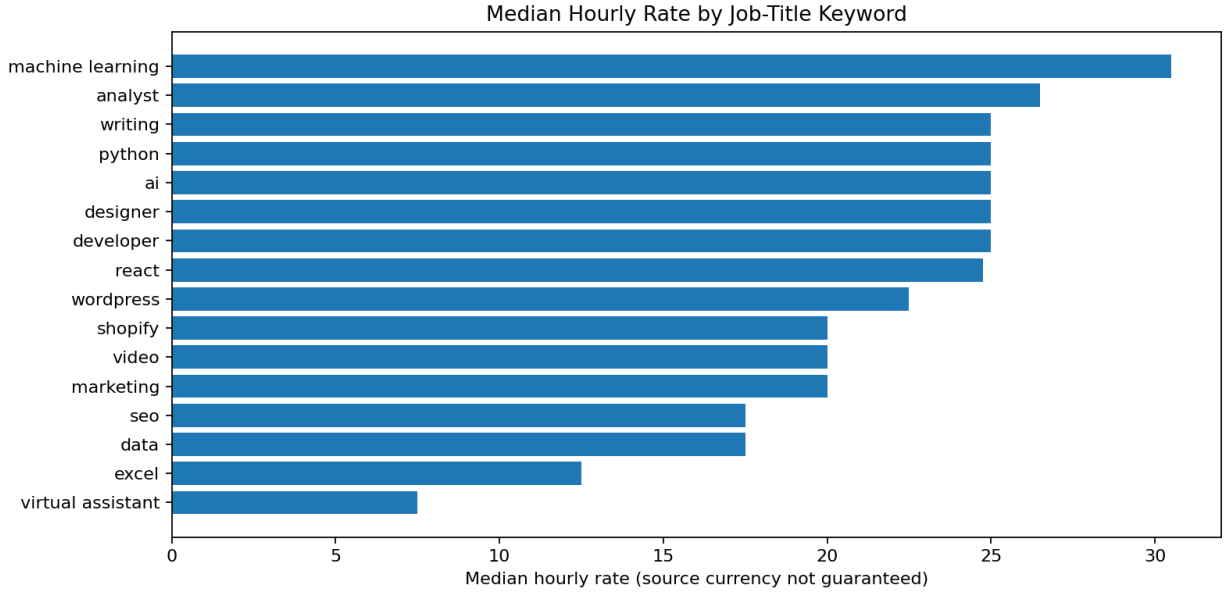
The dataset contains 244,828 freelance job postings. 140,937 postings are marked hourly and 103,891 are marked fixed-price/non-hourly. 102,422 records have enough hourly-rate information to calculate a midpoint. The project provides historical descriptive analysis and a transparent baseline recommendation app.

2. Data Quality and Interpretation

The date range is short and monthly counts are uneven, with most records in February and March 2024. Consequently, trend lines should not be interpreted as seasonality or a reliable long-term forecast. Compensation fields mix hourly rates and fixed budgets, so they are analyzed separately. Currency is not explicitly specified, which limits country-level interpretation.

3. Task 1 — Job Title Keywords and Compensation

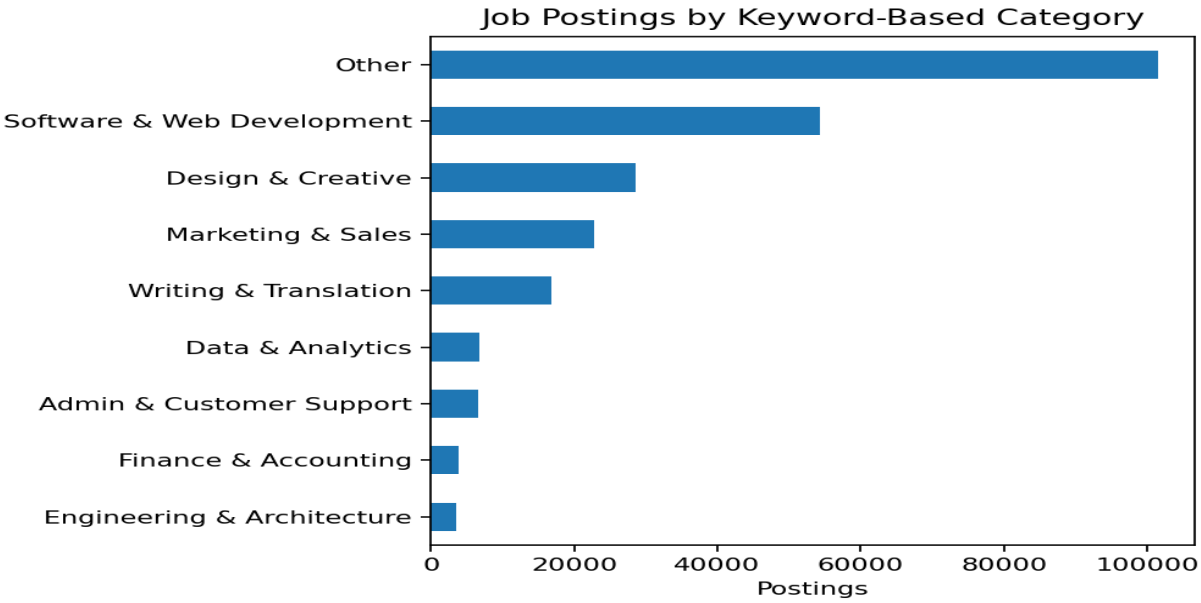
Titles are searched for a predefined set of skill/role keywords. For each keyword, the output table includes posting frequency, median hourly midpoint, and median fixed budget where available. Median is emphasized because compensation distributions can be skewed by unusually high values. These are associations, not evidence that a keyword causes higher pay.



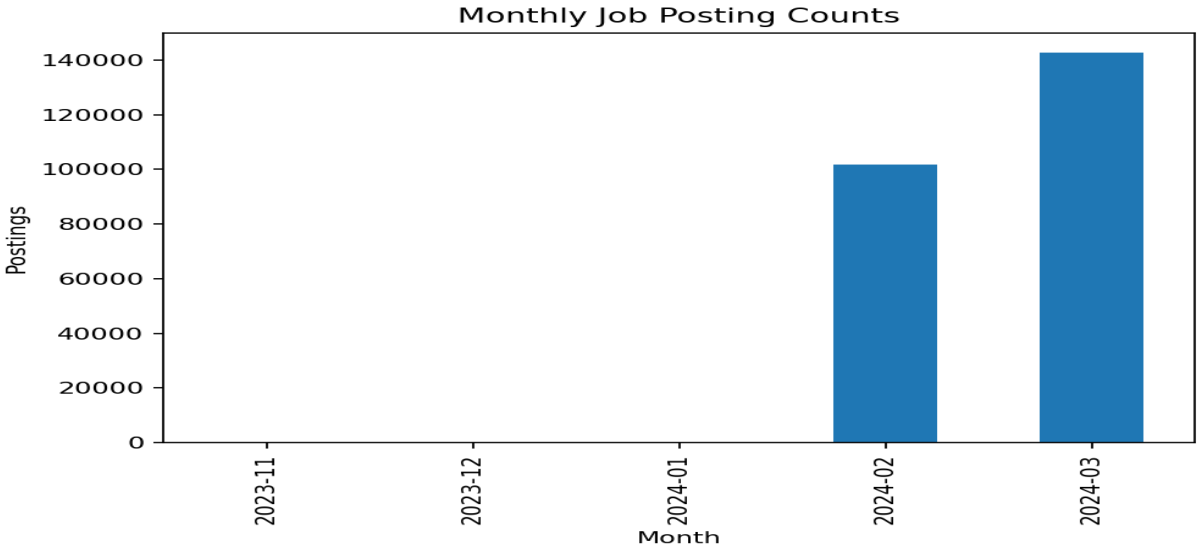
See outputs/title_keyword_salary_analysis.csv for the underlying keyword-level results.

4. Task 2 — Emerging Categories

Categories are heuristically assigned using title keywords. They provide a practical first pass for summarizing demand, but a job title may fit more than one category and the first matching rule is used. To claim a category is 'emerging', its frequency should increase over a sufficiently long, consistently sampled time series; this dataset alone is too short for strong claims.



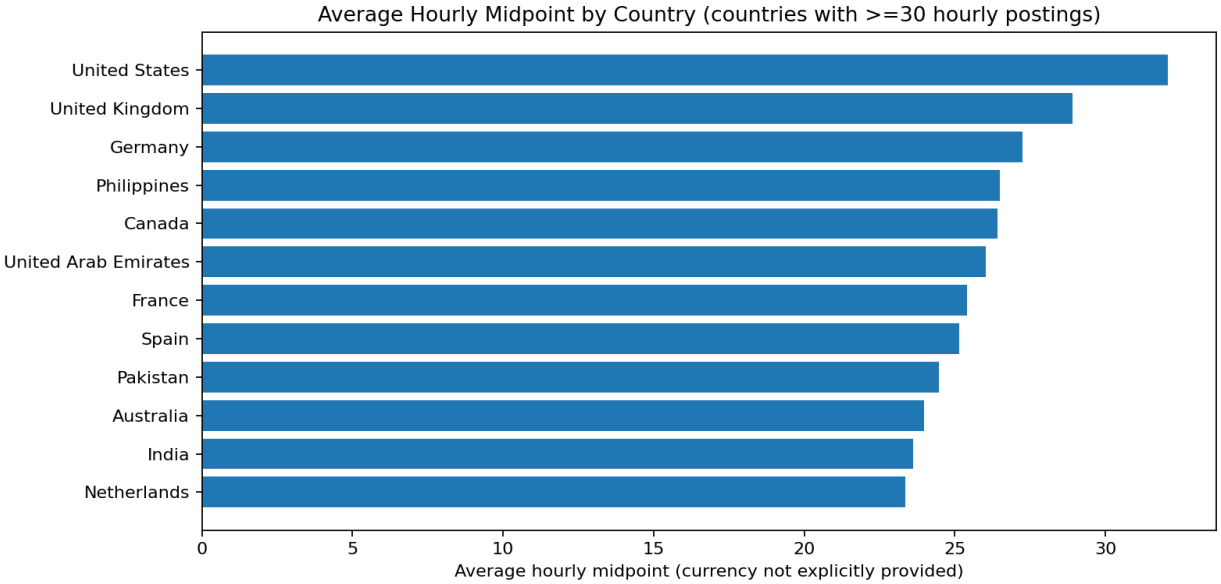
5. Task 3, 6 and 8 — Demand Trends and Forecasting



Monthly counts are delivered in outputs/monthly_trends.csv. A robust forecasting model should use longer history, validate on a time-ordered holdout set, and report MAE/RMSE against a simple baseline. This project intentionally avoids claiming high predictive accuracy from only a few months of highly uneven data.

6. Task 4 — Country Hourly Rate Comparison

Hourly midpoint is calculated as (hourly_low + hourly_high) / 2 where possible. Countries with at least 30 hourly postings are included in the exported comparison. Because currency metadata is absent, these figures are not guaranteed to be comparable purchasing-power or currency-adjusted wages.



See outputs/country_hourly_rates.csv for country-level counts, mean and median hourly midpoint.

7. Task 5 — Recommendation Engine

The Flask prototype lets users search title keywords, optionally filter by country, and optionally limit results to hourly jobs. It ranks postings by the number of query words matched in the title, then favors newer postings. It exposes GET /api/recommend and GET /health. This is an explainable baseline; a future version should use skills, seniority, location and user feedback and evaluate ranking quality.

8. Task 7 — Remote Work Landscape

The supplied schema does not contain a reliable remote/on-site field. A title-keyword proxy flags 5,694 postings containing remote, work from home, WFH, anywhere or virtual. This undercounts remote jobs whose titles omit those terms and may overcount roles mentioning virtual for other reasons.

9. Reproducibility and Deployment

Use the notebook for guided exploration, src/analysis.py for reproducible outputs, sql/schema_and_queries.sql for MySQL examples, and Dockerfile/docker-compose.yml for a containerized Flask app. Detailed commands are included in README.md.

10. Conclusion and Recommendations

The project provides a useful end-to-end portfolio baseline: data preparation, salary and country comparisons, category counts, time summaries, recommendation API and container setup. The highest-priority improvement is obtaining a longer, regularly refreshed dataset with explicit currency and work-mode fields. Only then should future-demand forecasts be presented as validated predictions.

Note: This report is based only on the user-supplied CSV and does not represent a live market survey.